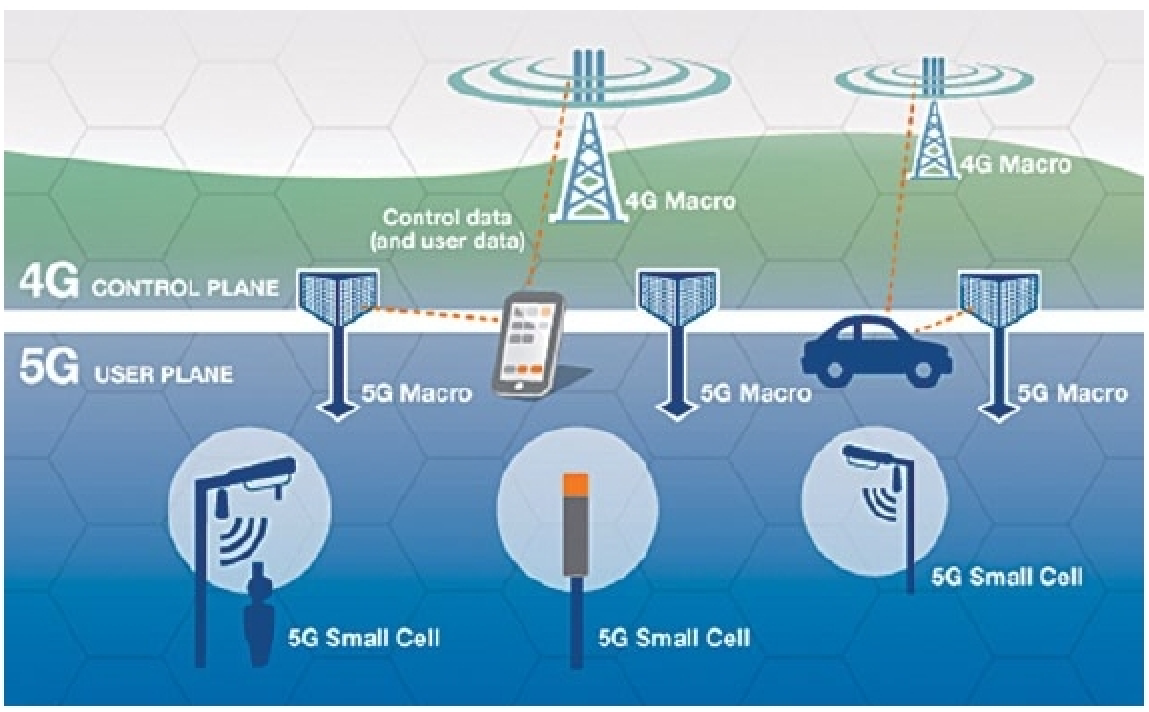
Advanced T&M Equipment For 5G And Their Demand

The number of trials for effective 5G implementation the world over, including India, has been on a constant rise. This is a key reason for increase in demand for 5G testers in the global market

Deepshikha Shukla

5G technology is unique in terms of its capabilities as compared to previous communication technologies. It features ultra-fast speed, low latency, large capacity, pervasive connectivity and high reliability. With the inevitability of 5G, such new markets as connected cars, autonomous driving, electric vehicles, mining, intelligent farming, remote telemedicine and virtual reality (VR) are emerging. By utilising the enhanced mobile broadband (eMBB) feature of 5G networks, companies can conduct artificial intelligence (AI)-based analytics for a large number of on-site videos captured by fixed and mobile cameras to facilitate precise and efficient operation and maintenance.

Chipset makers, network infrastructure companies, device OEMs and test and measurement (T&M) companies have been heavily involved in research and development of 5G for years. Now the work is coming to fruition as 5G devices will begin to hit the Indian market by 2020. Consumer smartphones and a variety of other devices, from 5G-backed routers and mobile hotspots to the Internet of Things (IoT) devices, will be seen in the market by next year. 5G will play a dual role, of the driver of new disruptive business models and the essential converger of four tech superpowers—mobile, cloud, AI and the IoT.



5G integration with 4G (Credit: www.quora.com)

Need for 5G test equipment

High-speed digital, wireless, aerospace and defense, and automotive companies need integrated active and passive components for such devices as cellphones, satellite communications and 5G base stations, to increase performance and reduce size of end products. These highly integrated devices require highly integrated test solutions that address radio frequency (RF) test challenges while providing advanced functionality and performance. However, testing methodologies have changed—what used to be measured on a single channel over a cable is now moving over-the-air.

Effective 5G testing is essential to optimise signal quality, network throughput and capacity. It is also important to ensure interoperability between user equipment and network infrastructure. To keep up with evolving standards and changing technology, network operators and equipment manufacturers need fast, flexible access to the latest T&M equipment.

Commercial rollout of sub-6GHz 5G networks has begun, and mmWave technology continues to be developed even as this rollout is underway. R&D teams around the globe have been tackling the challenges that are emerging with pre-